Citadel Securities, one of the world's leading HFT firms, executes approximately 35% of all U.S. listed retail volume. While their proprietary trading systems use multiple languages, Python is widely used for research and analysis within the firm.
Here's a simple conceptual example of what a high-frequency trading decision might look like in Python:

```python
# Simple concept example of a high-frequency trading decision
bid_price = 100.01  # Current highest buying price
ask_price = 100.03  # Current lowest selling price

# The spread is the difference between ask and bid
spread = ask_price - bid_price

# If the spread is wide enough, we might profit by buying and selling quickly
if spread > 0.02:
    print("Execute trade: Buy at bid, sell at ask")
else:
    print("Spread too narrow, don't trade")
```

Statistical Arbitrage

Statistical arbitrage strategies identify temporary price disparities between related securities and execute trades to profit from their eventual convergence. Python's data analysis capabilities make it ideal for identifying these opportunities.

A classic example is pairs trading, where two historically correlated stocks temporarily diverge in price. When AXP (American Express) and COF (Capital One Financial) diverged significantly during market turmoil in March 2020, algorithmic traders using cointegration analysis could identify this temporary anomaly and profit from their eventual reconvergence.

Here's a simplified example showing how Python can compare two related stocks:

```python
# Simple example of comparing two related stocks
apple_price = 180  # Price of Apple stock
microsoft_price = 350  # Price of Microsoft stock
```